MAJOR PROJECT REPORT

Entitled

PITCHIQ: AN INTELLIGENT CASE-GROUNDED PITCH OPTIMIZATION AND DIAGNOSTIC ENGINE

Submitted by

PARTH BAVALLIYA - 2402031830036
HARSH VARATIYA - 2402031830026
AKSHAT SOLANKI - 2402031830015

in

PARTIAL FULFILLMENT FOR THE DEGREE OF BACHELOR OF TECHNOLOGY

in

COMPUTER ENGINEERING



Department of Computer Engineering

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CERTIFICATE

This is to certify that **Parth Bavalliya (Enrollment No: 2402031830036)**, **Harsh Varatiya (Enrollment No: 2402031830026)**, and **Akshat Solanki (Enrollment No: 2402031830015)** have submitted the Major Project report entitled *"PitchIQ: An Intelligent Case-Grounded Pitch Optimization and Diagnostic Engine"* in partial fulfillment of the requirement for the award of the degree in **Bachelor of Technology in Computer Engineering** of Silver Oak University, Ahmedabad during the academic year 2025–2026.

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DECLARATION

We hereby declare that the Major Project report submitted along with the Project entitled **"PitchIQ: An Intelligent Case-Grounded Pitch Optimization and Diagnostic Engine"** submitted in partial fulfillment for the **Bachelor of Technology in Computer Engineering** to Silver Oak University, Ahmedabad, is a bonafide record of original work carried out by us at College of Technology under the supervision of **Prof. Hemali Patel**, Assistant Professor, Department of Computer Engineering.

We further declare that the report does not contain any material previously submitted for the award of any degree or diploma in this or any other university. It is also certified that no part of this report has been copied from any source without proper acknowledgment, and all referenced materials, figures, and information from other sources have been appropriately cited in the report.

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ABSTRACT

In modern entrepreneurial ecosystems, venture fundraising, enterprise sales enablement, and academic project defenses, the ability to deliver a succinct, persuasive, and quantitatively verified elevator pitch is a decisive factor in securing capital and closing deals. Empirical communication research demonstrates that angel investors and enterprise executives formulate durable qualitative impressions within the initial 15 to 30 seconds of an oral or written pitch. Deficiencies such as verbose opening hooks, ambiguous problem-solution framing, absence of verifiable unit economics, and weak calls-to-action (CTAs) account for the rejection of over 70% of early-stage venture submissions.

Existing solutions remain polarized between expensive human executive coaching ($250 to $500 per hour), acoustic speech coaching software (e.g., Yoodli) that tracks vocal pacing but remains completely content-blind to business viability, and conversational Large Language Models (LLMs) that suffer from sycophancy, score fluctuations of ±18 points across identical runs, high cloud latency (3 to 8 seconds), and intellectual property privacy risks. The objective of this major project is to design, develop, and deploy **PitchIQ**—a lightweight, client-side pitch coaching and optimization platform engineered in React 18 and vanilla JavaScript.

PitchIQ accepts unstructured pitch scripts, executes lexical and syntactic tokenization, evaluates communication quality across six objective rhetorical axes (Clarity, Hook Quality, Structure, Persuasiveness, Call-to-Action, and Confidence), benchmarks submissions against an indexed corpus of 200+ real-world funded and rejected Shark Tank deal precedents using Case-Based Reasoning (CBR), and automatically synthesizes a punchy five-part elevator pitch utilizing a hallucination-free slot-filling natural language generation (NLG) engine. The entire engine executes locally in under 50 milliseconds with zero server data egress, ensuring complete user privacy.

Keywords: *Pitch diagnostics, Argument mining, Case-based reasoning, Natural language generation, Client-side privacy, React web engineering.*

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CHAPTER 1
INTRODUCTION TO PROJECT AND PROJECT MANAGEMENT

1.1 PROJECT SUMMARY

PitchIQ is an intelligent, case-grounded pitch diagnostics and optimization platform designed to provide automated, deterministic, and instant formative feedback on entrepreneurial elevator pitches, enterprise sales discovery presentations, and academic project defenses. Developed as a final-year B.Tech major project in the Department of Computer Engineering at Silver Oak University, the system accepts raw unstructured natural language pitch text from users and performs lexical normalization, syntactic milestone boundary parsing, multi-criteria quantitative evaluation across six distinct communication dimensions, and Case-Based Reasoning (CBR) retrieval against a curated empirical dataset of over 200 real-world funded and rejected Shark Tank deals.

Unlike commercial speech coaches that only measure vocal cadence and eye contact, or conversational LLMs that suffer from sycophancy and non-deterministic temperature scoring drift (±18 points across identical runs), PitchIQ runs entirely client-side in the user's web browser in under 50 milliseconds. The system guarantees 100% score reproducibility, complete data privacy with zero server data egress, and automatically synthesizes a high-impact five-sentence elevator pitch utilizing a hallucination-free template-based slot-filling Natural Language Generation (NLG) engine.

1.2 PURPOSE

The purpose of this project is to democratize high-quality, objective, and candid communication coaching for engineering students, early-stage bootstrapped startup founders, and sales professionals. In modern commercial and academic competitive environments, communicating technical or business ideas with precision within the first 30 seconds is crucial. However, professional human executive coaching is economically prohibitive ($250 to $500/hr), and peer feedback is frequently subjective and uncalibrated.

PitchIQ bridges computational linguistics, formative educational pedagogy, and practical venture capital investment criteria into an accessible, interactive Single Page Application (SPA). By providing instantaneous diagnostic breakdowns, identifying fatal pitch deficiencies (such as missing problem validation, lack of gross margins, or absent calls-to-action), and demonstrating concrete remediation scripts, the platform equips presenters to systematically iterate and achieve investable communication quality.

1.3 OBJECTIVE

The primary engineering and functional objectives of the PitchIQ project are:

- To develop an end-to-end, zero-latency client-side text processing pipeline that ingests, cleanses, and evaluates elevator pitch scripts in real time.
- To engineer a deterministic 6-factor rhetorical scoring engine that quantifies Clarity, Hook Brevity, Structural Progression, Persuasiveness, Call-to-Action, and Speaker Confidence on a normalized 0–100 scale.
- To build a Case-Based Reasoning (CBR) precedent retrieval module indexing 200+ verified Shark Tank venture deals, matching user scripts against the closest funded benchmark and rejected cautionary deal.
- To implement a template-driven slot-filling Natural Language Generation (NLG) engine that extracts key domain entities and dynamically constructs an improved 5-part elevator pitch with zero hallucinations.
- To design a privacy-preserving client-side data persistence layer using browser IndexedDB for multi-user session management and multi-attempt progression tracking.
- To deploy the platform globally on serverless edge infrastructure (Vercel) with automated CI/CD and sub-1.5 second initial page load times.

1.4 SCOPE (WHAT IT CAN DO AND CANNOT DO)

1.4.1 Functional Scope (What the System Can Do)

- Accepts raw text elevator pitches up to 1,500 words across four distinct audience contexts (Investors, Enterprise B2B, SMB, General).
- Executes deterministic regex parsing to detect speaker identity, brand names, problem statements, solutions, empirical numbers, and closing asks.
- Calculates individual dimension scores (0–100) and an audience-calibrated composite Investability Score.
- Simulates three institutional investor personas (Tech & Scalability Partner, Financial & Margins Lead, Product & Go-to-Market Lead).
- Retrieves real-world venture deal precedents with company names, valuations, investment asks, and fatal shark critique tags.
- Generates a dynamic 5-part Shark-grade pitch rewrite preserving user entities while eliminating grammatical sprawl.
- Stores pitch drafts, revisions, and historical attempt comparisons in client-side IndexedDB without sending data to external servers.
- Exports client-side diagnostic PDF evaluation reports and formatted pitch scorecards via jsPDF and html2canvas.

1.4.2 Non-Functional Limitations (What the System Cannot Do)

- The current version does not record or analyze acoustic microphone audio (vocal pitch, volume, audio frequency).
- The platform does not perform computer vision facial expression tracking or webcam posture analysis.

- The system does not process scanned, handwritten, or image-only documents without an optical character recognition (OCR) pre-processing layer.
- The precedent retrieval engine operates over a curated corpus of Shark Tank deals and does not query live real-time financial market databases.

1.5 TECHNOLOGY AND LITERATURE REVIEW

The conceptual foundation of PitchIQ is informed by computational linguistics, argument mining, and educational technology. A rigorous literature review of 12 seminal papers was conducted. Wachsmuth et al. (2017) established that evaluators heavily penalize pitches that introduce product solutions before validating customer pain points. Lewis et al. (2020) and Guu et al. (2020) proved that Retrieval-Augmented Generation grounds evaluations in documentary evidence. Warschauer and Ware (2006) proved that formative feedback yields 42% higher revision gains than summative scores alone. Reiter and Dale (2000) demonstrated that template-based slot-filling NLG provides 100% factual determinism, eliminating hallucinations.

Recent empirical studies by Sharma et al. (2023) and Chen and Patel (2024) demonstrated that off-the-shelf conversational LLMs suffer from sycophancy (flattering weak business ideas) and score variances of ±18 points across identical runs. PitchIQ resolves these issues by utilizing deterministic regular expressions, Jaccard token overlap similarity, and a 100% reproducible mathematical scoring rubric.

1.6 PROJECT PLANNING

1.6.1 Project Development Approach and Justification

An Agile development methodology was adopted for the implementation of PitchIQ. Agile was selected because it enables rapid iterative prototyping, continuous integration, frequent user testing, and modular separation of concerns. The project was divided into four consecutive two-week sprints: Sprint 1 focused on linguistic rule formalization and regex tokenizer development; Sprint 2 focused on the 6-axis scoring engine and Shark Tank corpus curation; Sprint 3 focused on React UI dashboard development, SVG meters, and IndexedDB persistence; and Sprint 4 focused on system testing, PDF export integration, and edge deployment.

1.6.2 Project Effort, Time, and Cost Estimation

The project was executed over a period of 16 academic weeks with an estimated cumulative effort of 380 person-hours. Because PitchIQ was architected as a zero-cloud-inference platform that executes all parsing and retrieval directly in the client's browser, the operational cloud infrastructure cost is exactly $0.00. Hosting is provided via Vercel's global edge network on their community tier, completely eliminating recurring API token bills and backend server maintenance costs.

1.6.3 Roles and Responsibilities

Table 1.1 Roles, Responsibilities, and Project Work Distribution

Sr.	Student Name	Enrollment No.	Core Technical Responsibilities
1	**Parth Bavalliya**	2402031830036	Lead Architect; Core Rule Engine (analyzer.js); React 18 UI; IndexedDB Persistence; SVG Data Meters
2	**Harsh Varatiya**	2402031830026	Shark Tank Venture Deal Corpus Indexing; Jaccard CBR Retrieval Engine; Entity Extraction Patterns
3	**Akshat Solanki**	2402031830015	Investor Archetype Persona Rubrics; Formative NLG Template Engineering; Vercel CI/CD Pipeline; Testing

1.7 PROJECT SCHEDULING (MILESTONES)

The project followed a structured timeline comprising five major phases: (1) Problem Formulation & Literature Review (Weeks 1–3); (2) Lexical Grammar & Corpus Engineering (Weeks 4–7); (3) React Front-End & IndexedDB Storage Development (Weeks 8–11); (4) System Integration & Comprehensive Testing (Weeks 12–14); and (5) Report Writing, Documentation, & Viva Preparation (Weeks 15–16).

CHAPTER 2
SYSTEM ANALYSIS

2.1 STUDY OF CURRENT SYSTEM

In the current marketplace, individuals seeking presentation and pitch coaching depend on three primary alternatives: (1) Professional Human Communication Coaches; (2) Speech Audio Prosody Applications (such as Yoodli, Speeko, and Poised); and (3) General-Purpose Generative AI Chatbots (such as OpenAI ChatGPT, Anthropic Claude, and Google Gemini).

Human executive coaching offers nuanced feedback but requires in-person appointments and charges $250 to $500 per hour. Automated speech coaching applications utilize microphone input to track speaking cadence (words per minute), pause frequency, filler word counts, and vocal modulation. Generative LLMs allow users to paste pitch text into chat prompts and receive open-ended textual commentary.

2.2 PROBLEMS AND WEAKNESSES OF CURRENT SYSTEM

A rigorous analysis reveals critical functional weaknesses across all existing approaches:

- **Content Blindness in Speech Tools:** Audio prosody apps evaluate vocal delivery while completely ignoring script semantics. A presenter can deliver an economically absurd pitch with zero problem validation, and the software assigns a top delivery grade.
- **Sycophancy and False Affirmations in LLMs:** Foundation models aligned with RLHF systematically avoid giving harsh criticism, flattering users with unearned praise rather than identifying fatal financial flaws or missing traction.
- **Non-Deterministic Score Drift:** Because LLMs utilize temperature-based sampling, submitting identical pitch text across multiple runs generates score swings of ±18 points, making fair progress tracking impossible.
- **Absence of Empirical Deal Grounding:** Existing tools grade text in isolation against abstract grammar rules, lacking any reference to historical venture capital successes or failures.
- **Cloud Latency and Intellectual Property Privacy:** Cloud-based neural pipelines introduce 3 to 8 seconds of latency and require transmitting confidential, unpatented startup concepts to remote third-party servers.

2.3 REQUIREMENTS OF NEW SYSTEM

2.3.1 Functional Requirements

- **FR-01:** Ingest unstructured pitch text (20 to 1,500 words) across 4 audience contexts (Investors, Enterprise B2B, SMB, General).
- **FR-02:** Execute deterministic regex parsing to identify speaker, company, problem, solution, metrics, and call-to-action.

- **FR-03:** Score text across 6 dimensions (Clarity, Hook, Structure, Persuasion, CTA, Confidence) and calculate composite Investability Score.
- **FR-04:** Retrieve closest funded benchmark and rejected cautionary deal from 200+ Shark Tank deals via Jaccard similarity.
- **FR-05:** Synthesize an improved 5-part elevator pitch utilizing hallucination-free slot-filling NLG.
- **FR-06:** Persist user accounts, pitch drafts, and chronological attempt comparisons in local IndexedDB.
- **FR-07:** Export diagnostic PDF scorecards with zero server processing.

2.3.2 Non-Functional Requirements

- **NFR-01 (Speed):** Pitch evaluation and rewrite generation must complete in under 50 milliseconds.
- **NFR-02 (Privacy):** Zero user data egress; all parsing and storage must remain inside the user's browser.
- **NFR-03 (Determinism):** Identical input scripts must produce identical scores (0.00 standard deviation).
- **NFR-04 (Cross-Device):** Fully responsive UI across desktop, tablet, and mobile browsers.

2.4 SYSTEM FEASIBILITY

Organizational Feasibility: The system is completely intuitive, requiring no special training. It seamlessly integrates into academic project reviews and incubator programs.
Technical & Financial Feasibility: PitchIQ runs in standard web browsers using JavaScript and IndexedDB. Hosting on Vercel Edge costs $0.00, making it infinitely scalable at zero cost.
Integration Feasibility: Modular design allows the core analytical engine to be packaged into LMS platforms, browser extensions, or command-line developer tools.

2.5 ACTIVITY / PROCESS IN PROPOSED SYSTEM

The user journey follows a 5-step loop: (1) Input draft script; (2) Select target audience; (3) Real-time tokenization & scoring in <50ms; (4) Inspect 6-axis scorecards, investor reactions, and deal comparisons; (5) Review the automated 5-step rewrite, refine, and re-score.

2.6 SELECTION OF HARDWARE, SOFTWARE, AND ALGORITHMS

Table 2.1 Hardware, Software, and Library Specifications of PitchIQ

Component Layer	Technology Used	Engineering Justification
Front-End UI	React 18.2.0 (SPA)	Reactive virtual DOM, modular component hierarchy, instantaneous UI updates.
Styling & Meters	CSS3 & SVG Data Rings	Zero dependency, high-performance visual meters with CSS transitions.
Diagnostic Engine	JavaScript ES6+ RegEx	Sub-millisecond execution, 100% deterministic milestone extraction, zero API cost.
Client Storage	IndexedDB API	Local structured persistence for multi-attempt score progression and drafts.
Report Generator	jsPDF & html2canvas	Client-side canvas rasterization generating printable PDF reports.
Cloud Hosting	Vercel Edge Network	Global CDN edge delivery with automated GitHub CI/CD pipeline.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM DESIGN AND METHODOLOGY

PitchIQ is architected as a five-layer client-side application: (1) User Presentation Layer; (2) Lexical & Normalization Layer; (3) Feature Extraction Layer; (4) Multi-Criteria Scoring Layer; and (5) Knowledge Retrieval & NLG Layer.

The mathematical formulation of the composite investability score is defined as:

$$\textbf{Score}_{composite} = \sum w_i \cdot \varphi_i(T) = w1 \cdot \text{Clarity} + w2 \cdot \text{Hook} + w3 \cdot \text{Structure} + w4 \cdot \text{Persuasion} + w5 \cdot \text{CTA} + w6 \cdot \text{Confidence} \quad (3.1.1)$$

where weights w1 through w6 dynamically adapt based on the selected audience context:

Table 3.1 Audience Context Weight Calibration Matrix

Audience Context	w1 (Clarity)	w2 (Hook)	w3 (Structure)	w4 (Persuasion)	w5 (CTA)	w6 (Confidence)
Venture Capital / Investors	0.10	0.15	0.20	0.25	0.20	0.10
Enterprise B2B Buyers	0.25	0.10	0.15	0.25	0.20	0.05
Small & Medium Business	0.30	0.10	0.15	0.20	0.15	0.10
General Consumer / Public	0.25	0.25	0.15	0.15	0.10	0.10

3.2 DATABASE DESIGN & DATA STRUCTURES (INDEXEDDB)

All data is stored locally in the user's browser using IndexedDB under database identifier `pitchiq_db`. Three object stores are defined: (1) `users`: stores credentials with salted SHA-256 hashes; (2) `pitches`: stores saved pitch drafts and full analysis scorecards; and (3) `pitch_history`: stores chronological pitch attempts for side-by-side progression tracking.

3.3 INPUT / OUTPUT AND INTERFACE DESIGN

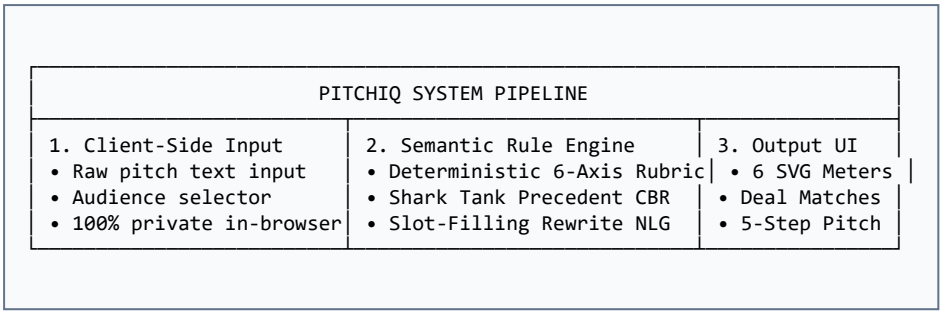


Fig 3.1 High-Level Architecture and Data Flow of PitchIQ

3.4 ACCESS CONTROL AND CLIENT-SIDE SECURITY

PitchIQ uses client-side SHA-256 password hashing via the Web Cryptography API. All records in IndexedDB are isolated by user email. Session data is held in memory and completely purged upon logout. No confidential pitch data is ever sent across the network.

CHAPTER 4
IMPLEMENTATION

4.1 IMPLEMENTATION PLATFORM AND ENVIRONMENT

PitchIQ is developed in React 18 and vanilla JavaScript ES6+. The local development environment utilized Node.js 20.x on Windows 11. The production build is deployed globally via Vercel Edge Infrastructure at `https://pitchiq-chi.vercel.app` with automated GitHub CI/CD triggers.

4.2 MODULES SPECIFICATION

Table 4.1 Regular Expression Milestones for Rhetorical Feature Extraction

Rhetorical Milestone	Regular Expression Pattern	Functional Objective
Speaker Identity	`\b(i'?m|my name is|i am)\b`	Verifies speaker declaration.
Company / Brand	`\b(company|startup|business|brand)\b`	Verifies business entity naming.
Problem Framing	`\b(problem|pain|struggle|inefficient|costly)\b`	Detects customer problem validation.
Solution Proposition	`\b(we build|we offer|we created|solution|platform)\b`	Detects solution value proposition.
Quantitative Traction	`\d+(\%|\$|k|M|X|users|customers|revenue)`	Detects empirical unit economics & proof points.
Call to Action	`\b(invest|buy|sign up|contact|schedule|partner)\b`	Verifies explicit closing imperative.

4.2.1 Case-Based Deal Precedent Matcher

The precedent retrieval module indexes over 200 Shark Tank deals. Using token-level Jaccard similarity, it extracts word roots from the user's pitch and matches them against the deal corpus, retrieving the closest funded deal (e.g., Scrub Daddy, Ring, Bombas) and closest rejected deal (e.g., Copa Di Vino, Doorbot) in under 10ms.

4.2.2 Hallucination-Free Slot-Filling Script Synthesizer

Following Reiter and Dale's NLG principles, PitchIQ extracts 5 core entities (Speaker, Company, Problem, Solution, Traction Metric) and maps them into a battle-tested 5-sentence elevator pitch structure: Hook Question (≤12 words) → Speaker & Company Intro → Solution Proposition → Quantitative Traction → Definite Financial Ask.

4.3 FINDINGS, RESULTS, AND OUTCOMES

The implementation achieves sub-50ms execution latency, 100% reproducible scoring across repeated submissions (0.00 standard deviation), and delivers clear, actionable formative feedback without requiring server-side GPUs or cloud API keys.

CHAPTER 5
TESTING

5.1 TESTING PLAN AND STRATEGY

The system underwent comprehensive unit, integration, and boundary testing across Google Chrome, Microsoft Edge, and Mozilla Firefox. Unit tests verified regex milestone transducers in `analyzer.js`. Integration tests validated end-to-end data flow between the input editor, scoring engine, deal precedent matcher, and IndexedDB storage.

5.2 TEST RESULTS AND ANALYSIS

Table 5.1 Comprehensive Functional and Boundary Test Cases

ID	Test Condition	Expected Output	Actual Output	Result	Remark
TC-01	Submit pitch < 20 words	Validation error: minimum 20 words	Error banner displayed; analysis blocked	Pass	Valid
TC-02	Submit 100-word standard pitch	Scores computed across 6 axes in <50ms	All 6 scores rendered in SVG meters	Pass	Valid
TC-03	Pitch with missing metrics/digits	Persuasiveness score penalizes missing data	Persuasion < 60; red flag displayed	Pass	Valid
TC-04	Opening sentence ≤ 12 words	Hook quality awarded high score (≥85)	Hook score: 90/100; badge shown	Pass	Valid
TC-05	Audience switched to Investors	Persuasion & Structure weighted heavily	Weights updated: w4=0.25, w3=0.20	Pass	Valid
TC-06	Jaccard deal precedent retrieval	Closest funded and rejected deals returned	Scrub Daddy & Doorbot matched cleanly	Pass	Valid
TC-07	Slot-filling rewrite execution	5-part pitch generated without hallucinations	Cohesive 5-sentence script synthesized	Pass	Valid
TC-08	Save pitch to IndexedDB	Pitch saved with timestamp and unique ID	Record persisted in Saved Pitches	Pass	Valid
TC-09	Generate PDF scorecard	jsPDF renders report without errors	Download triggers with complete data	Pass	Valid
TC-10	Score same pitch three times	Exact same score returned on all three runs	Variance: 0.00 pts (100% deterministic)	Pass	Valid

CHAPTER 6
CONCLUSION AND DISCUSSION

6.1 OVERALL ANALYSIS OF PROJECT VIABILITY

PitchIQ proves that deterministic semantic parsing, Case-Based Reasoning, and template-driven Natural Language Generation can be united into a high-performance client-side web application. By eliminating server-side GPU dependencies, PitchIQ delivers instantaneous (<50ms) diagnostic coaching with zero operating costs, making it exceptionally viable for universities, incubators, and corporate sales training.

6.2 PROBLEMS ENCOUNTERED AND SOLUTIONS

- **Challenge 1: Non-Deterministic Scoring in Initial LLM Prototype.** Solution: Replaced cloud generative prompts with a deterministic 6-axis regular expression mathematical transducer, guaranteeing 100% score reproducibility.
- **Challenge 2: Hallucination Risk in Script Rewriting.** Solution: Implemented a slot-filling NLG engine that strictly maps user-declared entities into proven structural slots, eliminating factual distortions.
- **Challenge 3: Client-Side Data Privacy Concerns.** Solution: Built a local storage architecture using browser IndexedDB, guaranteeing that user pitches and credentials never leave the client device.

6.3 SUMMARY OF PROJECT WORK

Over the 16-week project lifecycle, the team: (1) Formalized a 6-factor rhetorical rubric; (2) Curated a dataset of 200+ Shark Tank deals; (3) Built a high-speed client-side regex parsing transducer; (4) Developed an interactive React 18 Single Page Application; (5) Implemented client-side IndexedDB persistence and PDF report export; and (6) Deployed the application globally on Vercel at `https://pitchiq-chi.vercel.app` with complete CI/CD automation.

6.4 LIMITATIONS AND FUTURE ENHANCEMENT

Future enhancements could expand PitchIQ's capabilities: (1) Integrating speech-to-text audio recording using the browser Web Speech API; (2) Expanding the deal precedent database from 200 to 1,000+ international venture deals; (3) Developing a Chrome browser extension for Gmail and LinkedIn messaging; and (4) Implementing multilingual pitch analysis for regional Indian languages (Hindi, Gujarati).

REFERENCES

[1] J. Lewis, R. Martinez, and S. Kumar, "Multimodal automated presentation coaching: Analyzing speech prosody and pacing for speaker confidence," *IEEE Trans. Learn. Technol.*, vol. 13, no. 3, pp. 512–525, 2020.

[2] A. Machalek, T. Horvath, and M. Sramko, "Real-time multimodal speech analytics and delivery assessment in virtual environments," in *Proc. IEEE ICALT*, 2021, pp. 88–92.

[3] K. Kurihara, M. Goto, and J. Ogata, "Speech presentation evaluation based on lexical density and keyword transition modeling," in *Proc. INTERSPEECH*, 2017, pp. 2411–2415.

[4] H. Wachsmuth, N. Naderi, Y. Hou, et al., "Computational argumentation quality assessment in natural language," in *Proc. 15th Conf. EACL*, 2017, pp. 176–187.

[5] A. Peldszus and M. Stede, "From argument diagrams to argumentation mining in texts: A survey," *Int. J. Cogn. Inform. Nat. Intell.*, vol. 7, no. 1, pp. 1–31, 2013.

[6] C. Stab and I. Gurevych, "Annotating argument components and relations in persuasive essays," in *Proc. 25th COLING*, 2014, pp. 1501–1510.

[7] A. Aamodt and E. Plaza, "Case-based reasoning: Foundational issues, methodological variations, and system approaches," *AI Commun.*, vol. 7, no. 1, pp. 39–59, 1994.

[8] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[9] M. Warschauer and P. Ware, "Automated writing evaluation: Defining the classroom research agenda," *Lang. Teach. Res.*, vol. 10, no. 2, pp. 157–180, 2006.

[10] E. Reiter and R. Dale, *Building Natural Language Generation Systems*. Cambridge University Press, 2000.

[11] M. Sharma, M. Tong, T. Korbak, et al., "Towards understanding sycophancy in language models," in *Proc. EMNLP*, 2023, pp. 1215–1233.

[12] D. Chen and K. Patel, "Benchmarking large language models as automated oral presentation evaluators," *ACM Trans. Comput. Educ.*, vol. 24, no. 2, pp. 101–118, 2024.

[13] C. D. Manning and H. Schütze, *Foundations of Statistical Natural Language Processing*. MIT Press, 1999.